

Problem 2.3

Solution:

(i) Subproblems:

For each $i = 1, \dots, n$, define

$P(i, s_{1_i}, s_{2_i})$ = minimum total waiting time when
 each server's serving time is $s_{1_i}, s_{2_i}, s_{3_i}$
 after customer i ,

subject to

$$s_{1_i} + s_{2_i} + s_{3_i} = \sum_{k=0}^i \ell_k.$$

Want

$$\min_{s_1, s_2 \in (0, \dots, nL)} P(n, s_1, s_2).$$

(ii) Recursive formula:

For all $i = 1, \dots, n$,

$$P(i, s_{1_i} + \ell_i, s_{2_i}) = \min\{P(i, s_{1_i}, s_{2_i}), P(i-1, s_{1_i}, s_{2_i}) + s_{1_i}\},$$

$$P(i, s_{1_i}, s_{2_i} + \ell_i) = \min\{P(i, s_{1_i}, s_{2_i}), P(i-1, s_{1_i}, s_{2_i}) + s_{2_i}\},$$

$$P(i, s_{1_i}, s_{2_i}) = \min\{P(i, s_{1_i}, s_{2_i}), P(i-1, s_{1_i}, s_{2_i}) + s_{3_i}\},$$

where

$$s_{3_i} = \sum_{k=0}^i \ell_k - s_{1_i} - s_{2_i}.$$

Base case is $P(0, 0, 0) = 0$.

(iii) Evaluation order:

Evaluate i from 1 to n , evaluate s_1, s_2 from 0 to nL , for each (i, s_1, s_2) , evaluate $P(i, s_{1_i} + \ell_i, s_{2_i})$, $P(i, s_{1_i}, s_{2_i} + \ell_i)$ and $P(i, s_{1_i}, s_{2_i})$ iff $(i-1, s_{1_i}, s_{2_i})$ is reachable.

(iv) Optimal value:

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L = pre-computed sum(\ell_i)
P = array of size (2, nL, nL) initiated with +inf
S = array of size (n+1, nL, nL) of previous state (s1, s2)
P[0][0][0] = 0
for i = 1 to n:
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roll P
for s1 = 0 to n*L:
  for s2 = 0 to n*L:
    if P[i-1][s1][s2] != +inf: # previous state is reachable
      evaluate P according to step (ii)
      update S
return minnum value of P[n]
      optimal state if optimal solution is needed

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(v) Optimal solution:

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state = optimal state of n (s1_n, s2_n)
assignment = [0]*n
for i = n to 1:
  previous_state = S[k_i][s1_i][s2_i]
  # get assignment by comparing server time
  assignment[i] = k of any sk whose serve time is different
                  between the states, k=3 otherwise
  state = previous_state
return assignment

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(vi) Time and space analysis:

Iteration takes $O(n \times (nL)^2)$ steps, each step takes $O(1)$ time to evaluate P and record state and assignment S , optimal solution takes extra $O(n)$ time, total time complexity is $O(n^3L^2)$. P takes $O((nL)^2)$ space using rolling array, S takes $O(n \times (nL)^2)$ space, total space complexity is $O(n^3L^2)$. If optimal solution is not needed, total space complexity is $O(n^2L^2)$.